

Temperature-Dependent Constrained Diffusion of Micro-Confined Alkylimidazolium Chloride Ionic Liquids

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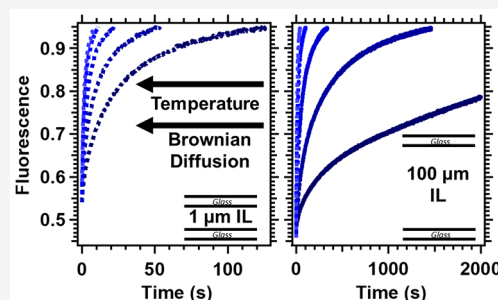


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ABSTRACT: Alkylimidazolium chloride ionic liquids (ILs) have many uses in a variety of separation systems, including micro-confined separation systems. To understand the separation mechanism in these systems, the diffusion properties of analytes in ILs under relevant operating conditions, including micro-confinement dimension and temperature, should be known. For example, separation efficiencies for various IL-based microextraction techniques are dependent on the sample volume and temperature. Temperature-dependent (20–100 °C) fluorescence recovery after photobleaching (FRAP) was utilized to determine the diffusion properties of a zwitterionic, hydrophilic dye, ATTO 647, in alkylimidazolium chloride ILs in micro-confined geometries. These micro-confined geometries were generated by sandwiching the IL between glass substrates that were separated by ~1 to 100 μm . From the measured temperature-dependent FRAP data, we note alkyl chain length-, thickness-, and temperature-dependent diffusion coefficients, with values ranging from 0.021 to 46 $\mu\text{m}^2/\text{s}$. Deviations from Brownian diffusion are observed at lower temperatures and increasingly less so at elevated temperatures; the differences are attributed to alterations in intermolecular interactions that reduce temperature-dependent nanoscale structural heterogeneities. The temperature- and thickness-dependent data provide a useful foundation for efficient design of micro-confined IL separation systems.



INTRODUCTION

Ionic liquids (ILs) are traditionally defined as organic salts that have a melting point below 100 °C.^{1–3} They are generally composed of an organic cation paired with either an organic or inorganic anion and are known to have unique physiochemical properties that make them particularly useful solvents. Among these properties, ILs customarily have low-to-negligible vapor pressure, wide solvating power, and sufficient thermal stability.^{4–6} By choosing the appropriate cation or anion, the properties of an IL can be tuned to fit specific applications. This extraordinary ability to uniquely design an IL for a specific task has attracted considerable attention for the extraction and separation of organic compounds, metal ions, biomolecules, and other compounds. ILs have been used as solvents for dispersive liquid–liquid microextraction,^{7–11} single-drop microextraction,^{12–14} and solid-phase microextraction,¹⁵ among other related microextraction techniques in which ILs are confined either in micrometer-sized sample volumes or within ~200 μm distances from a solid support.

ILs are often characterized by their high degree of structural heterogeneity that occurs over the spatial scale of a few nanometers.^{16–20} The nanoscale structural heterogeneity is experimentally attributed to the segregation of alkyl tails of the cation into mesoscopic domains.²¹ The nanoscale structural heterogeneity of ILs has been linked to many properties, including their ability to dissolve polar and non-polar

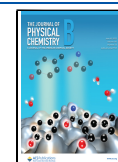
compounds,^{22–25} the viscosity trend observed upon increasing the alkyl chain length,²⁶ and other complex solvation dynamics.^{27–29} Margulis et al. noted that the highly ordered nature and viscosity of ILs is closely linked to their non-Newtonian behavior and the existence of nanoscale heterogeneous structures in ILs.^{30,31} In a study of the effect of increasing alkyl chain length in alkylimidazolium nitrate using coarse-grained molecular dynamics simulations, Wang et al. noted a structural transition from spatially heterogeneous to crystal-like with alkyl chain lengths above 14 carbons due to increased van der Waals interactions.³² However, from $n = 6$ to $n = 14$ carbons, they noted that the van der Waals interactions between the alkyl side chains were strong, but still weaker than the electrostatic interactions of the cation polar head groups; these polar groups form a continuous network that force the nonpolar alkyl chains to form separate domains.^{32,33}

Diffusion plays a role in the mechanism for many extraction and separation systems. It is imperative to understand the factors that may influence the diffusion of analytes in these

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systems. Several experimental and computational studies have been conducted to understand diffusion dynamics of analytes within bulk ILs (i.e., located far enough away from a surface so as not to be influenced by it) and ILs in micro-confined environments (i.e., films confined to 1 to 200 μm away from a solid substrate or sandwiched between two substrates).^{21,34–37} As evidenced by fluorescence correlation spectroscopy studies, the nanoscale structural heterogeneity in ILs can cause non-Brownian diffusion of small molecules through the ILs.³⁸ In addition, using temperature-dependent fluorescence lifetime spectroscopy, Welton et al. determined a link between the nanoscale structural heterogeneity and the observed local viscosity of the cybotactic region in ILs, which can alter the diffusion properties of small molecules.³⁹ Depending on the solvent and solute, negative and positive deviations from the Stokes–Einstein relationship, as well as agreement with the Stokes–Einstein relationship, have been reported.^{40–43} In pyrrolidinium and alkyl-phosphonium-based ILs, Maroncelli et al. described a charge-dependent solute diffusion that deviated from the Stokes–Einstein relationship; the characteristics of diffusion were attributed to solute size/shape as well as subtle IL packing effects and intermolecular interactions.⁴⁴

ILs near a solid surface (i.e., from a nanometer to approximately 1 μm from the surface) have been shown to behave markedly different due to long-range ordering compared to ILs in bulk phases, although the degree and extent of this ordering is dependent on the IL.^{45–48} The layered structure that defines long-range ordering is attributed to intermolecular interactions that are a combination of IL–substrate and IL–IL interactions.⁴⁹ While Shaw et al. demonstrated long-range ordering that extended several microns away from a solid polycrystalline silver support for alkylimidazolium bis(trifluoromethylsulfonyl)imide ILs,⁵⁰ Marzael et al. noted long-range ordering of confined ($\sim 1\ \mu\text{m}$) 1-hexyl-3-methylimidazolium ethylsulfate extending up to ~ 50 nm away from the solid mica support.⁴⁹ In ethylammonium nitrate, utilizing pulsed-field gradient nuclear magnetic resonance diffusometry, Fillippov et al. noted accelerated self-diffusion coefficients (compared with that in the bulk) of the ethylammonium nitrate IL confined between micron-separated glass plates.⁵¹ This acceleration was attributed to weakening of the hydrogen bond network seen in bulk ILs and was found to be independent of the substrate surface morphology and chemistry.^{52,53}

Many temperature-dependent physical and chemical properties of ILs are dependent on alkyl chain length and anion/cation pairing, including glass transition temperature, thermal stability, viscosity, density, and refractive index.^{54,55} The temperature-dependent properties of ILs are typically attributed to variations in intermolecular interactions, with many reports noting the important role hydrogen bonding plays within the IL network.^{56,57} Using neutron diffraction and neutron quasi-elastic scattering measurements, Nagao et al. noted shortening of the inter-alkyl distance of the imidazolium cation in 1-methyl-3-octylimidazolium chloride with decreasing temperature; thus, the alkyl chain becomes more compact with decreasing temperature, leading to an increase in the ordering of the nanoscale structure.⁵⁸ This temperature-dependent nanoscale structural change is seen only with longer alkyl chain imidazolium ILs ($n > 4$).⁵⁶ The size of nanoscale structural heterogeneities in ILs varies with increasing alkyl chain length. Triolo et al. noted non-polar domain sizes at room temperature varying from 1.3 to 2.7 nm

when increasing the alkyl chain length from $n = 3$ to $n = 10$ in alkylimidazolium chloride ILs (with an increase of 0.2 nm per $-\text{CH}_2-$ unit).¹⁶ Temperature-dependent IL structures can be altered in interfacial regions near a solid support. Atkin et al. observed a decrease in the number of solvation layers in alkyl ammonium ILs formed on a mica surface when increasing the temperature of the system from 14 to 30 $^{\circ}\text{C}$; this temperature dependency was attributed to the weakening of the hydrogen bonding network.⁵⁹

We aim to provide the temperature-dependent diffusion properties of a small molecule, ATTO 647, in micro-confined ILs. We utilize FRAP to gain insight into the temperature dependence and film thickness dependence of diffusion in imidazolium-based ILs. FRAP has been utilized to measure diffusion properties within polymer films/solutions,^{60–63} multilayer films,^{64,65} hydrogels,^{66–69} self-assembled monolayers,⁷⁰ and ILs.^{71,72} For example, Samanta et al. utilized FRAP to monitor the diffusion of 4-(*N,N*-dimethylamino) benzonitrile in 1-butyl-3-methylimidazolium hexafluorophosphate; it was noted that the experimentally determined diffusion coefficient was a factor of 7 smaller than that predicted by the Stokes–Einstein relationship.⁷¹ The diffusion properties reported herein of ATTO 647 in 1-methyl-3-hexylimidazolium chloride, $[\text{C}_6\text{mim}][\text{Cl}]$, and 1-methyl-3-octylimidazolium chloride, $[\text{C}_8\text{mim}][\text{Cl}]$, can be utilized to better design IL-based separation systems for improved efficiencies.

EXPERIMENTAL SECTION

Sample Preparation. Figure 1 depicts the structures of the ILs, glycerol, and fluorophore utilized for this study. Glycerol

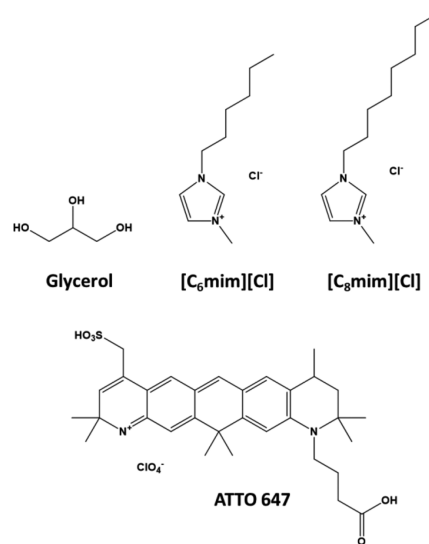


Figure 1. Structure of solvents and fluorophore utilized in this study.

($\geq 99.5\%$), $[\text{C}_6\text{mim}][\text{Cl}]$ ($\geq 97.0\%$), and $[\text{C}_8\text{mim}][\text{Cl}]$ ($\geq 97.0\%$) were purchased from Sigma-Aldrich (Milwaukee, WI), and ATTO 647 was purchased from ATTO-Tec GmbH (Siegen, Germany). ATTO 647 was chosen as the fluorophore for this study, as it has an appropriate excitation wavelength for the available laser (see FRAP Data Collection for further details), is photostable, and is commercially available. Solvation in ILs is a complicated subject,^{27–29} and no interpretation is made here as to the nature of the fluorophores' solvation environment.

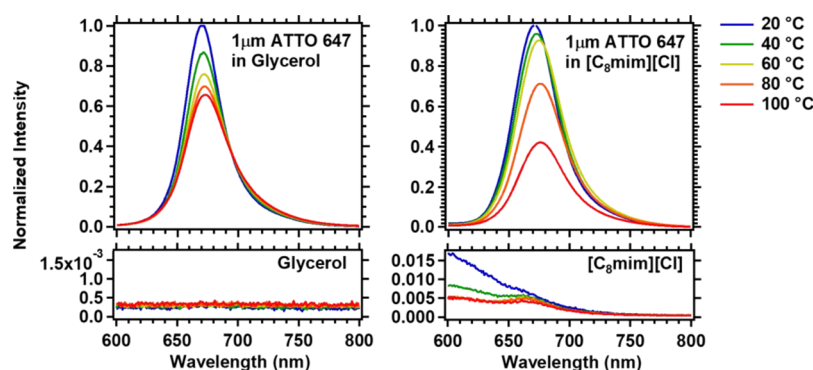


Figure 2. Temperature-dependent fluorescence emission spectra (580 nm excitation) of (upper panels) 1 μ M ATTO 647 in both glycerol and $[\text{C}_8\text{mim}][\text{Cl}]$ and (lower panels) glycerol and $[\text{C}_8\text{mim}][\text{Cl}]$ in the absence of fluorophore. A low intensity peak was detected at 668 nm in $[\text{C}_8\text{mim}][\text{Cl}]$; the emission peak is assumed to be an impurity.

Prior to use, glycerol and ILs (i.e., solvents) were dried under vacuum (-30 mmHg) for 24–72 h at room temperature (20 – 22 $^{\circ}\text{C}$) to remove excess water. After drying, a DL39 Karl Fischer Coulometric Titrator (Mettler Toledo, Columbus, OH) was utilized to confirm <100 ppm water in the ILs and glycerol. Temperature-dependent (20 – 100 $^{\circ}\text{C}$) viscosity measurements of the dry ILs were performed utilizing a DV1 Brookfield digital viscometer with a CPE-51 spindle, equipped with a ColePalmer Circulating Bath with a Standard Digital Temperature Controller (Polyscience, Niles, IL) to regulate temperature of the system.

For the fluorescence studies, an appropriate quantity of stock ATTO 647 in methanol (Sigma-Aldrich) was added to the previously dried solvent to reach a final concentration of 1 μ M. Fluorescence emission spectra were measured utilizing a HORIBA Fluorolog-3 spectrofluorometer (HORIBA Scientific, Edison, NJ) with 580 nm excitation, 5 s integration, and 1 and 5 nm excitation and emission slit width, respectively. As expected, the fluorescence intensity decreased with increasing temperature (Figure 2). This ATTO 647 concentration provided a sufficient signal over the entire temperature range that was utilized for this study; in addition, this concentration is below previously reported work that stated negligible probe effects on the IL structure.^{73,74} The temperature-dependent fluorescence spectra of glycerol and $[\text{C}_8\text{mim}][\text{Cl}]$ were also collected in the absence of ATTO 647. For all temperatures, the signal in the absence of fluorophore at this excitation wavelength was at most 1.5% ($[\text{C}_8\text{mim}][\text{Cl}]$) or 0.07% (glycerol) of the total signal obtained upon the addition of fluorophore. After the addition of the fluorophore, ILs and glycerol were dried under vacuum for 24 h at room temperature (20 $^{\circ}\text{C}$) to remove methanol.

All samples were prepared on piranha-treated glass. In short, glass substrates were sonicated in piranha solution (3:1, concentrated sulfuric acid:30% hydrogen peroxide solution) for 30 min at room temperature. **Caution!** Piranha solution is a strong oxidizing agent, which reacts violently with most organic materials and must be handled with extreme care. Following piranha treatment, the substrates were rinsed and stored in deionized water until use.

Figure S1 illustrates the three methods utilized to create the IL films. To prepare spin-coated films, 18 mm² coverslips (no. 1.5 Corning Inc., Corning, NY) were placed on a small chuck in a KW-4A spin coater (Chemat Technology Inc., Northridge, CA). Approximately 100 μL of fluorophore containing ILs or glycerol were drop-cast onto the surface of the glass,

immediately followed by rotation of the glass at 8000 rpm for 1 min. Following spin-coating, coverslips were then inverted and placed on glass slides and the edge-sealed with epoxy. For simplicity, films prepared using spin-coating will be referred to as the 1 μm thick films. To prepare the 10 μm thick films, approximately 5 μL of fluorophore containing solvent was drop-cast onto the surface of the glass slide and a coverslip was immediately placed on top. The solvent was allowed to spread to a uniform thickness for 15 min prior to sealing the edges of the coverslip with epoxy (Clearweld, JB Weld, Sulphur Springs, TX). To prepare the 100 μm thick films, 9 $\mu\text{m} \times 120$ μm circular SecureSeal imaging spacer (Grace Bio-Labs, Bend, OR) was placed on a glass slide. Approximately 8 μL of fluorophore containing solvent was placed inside the well and a coverslip was adhered to the top surface. As with the aforementioned sample geometries, samples were edge-sealed with epoxy to maintain sample uniformity. All samples were prepared in a nitrogen gas-filled isolation glove box ($<10\%$ humidity) (Cleatech, Orange, CA) to avoid water sorption from the environment during sample preparation. All measured film thicknesses are reported in Table 1. Thickness measure-

Table 1. Film Thicknesses^a

name	glycerol	$[\text{C}_6\text{mim}][\text{Cl}]$	$[\text{C}_8\text{mim}][\text{Cl}]$
1 μm film (spin-coated)	0.8 μm	1.2 μm	1.5 μm
10 μm film	12.1 μm	13.6 μm	14.9 μm
100 μm film	120.0 μm	119.7 μm	121.2 μm

^aThickness measurements were performed on samples prepared in the same manner as the FRAP samples; however, the fluorophore was not included in these samples. Measurements were conducted at room temperature (~ 20 $^{\circ}\text{C}$).

ments (Table 1) were conducted optically using an inverted confocal microscope (Nikon, Eclipse Ti2) equipped with a 100×1.49 NA objective and piezo electric stage with 0.2 nm axial precision (Physik Instrumente, Auburn, MA).

FRAP Data Collection. The FRAP instrument was based on an upright confocal microscope (Nikon, 80i) equipped with an OBIS 637-nm continuous-wave diode laser (Coherent, Santa Clara, CA) as the bleaching beam and a mercury lamp (Excelitas Technologies, Mississauga, Ontario) filtered with a long-pass filter (635 nm EdgeBasic, Semrock, Rochester, NY) as the imaging beam. The solvents in this study negligibly absorb in the red region; thus, a 637 nm laser was selected to bleach only the fluorophore, ATTO 647 (Figure S2). The

fluorescence signal was collected with a 10× objective (0.3 NA, Nikon, Melville, NY) and filtered through a dichroic mirror (BrightLine, Semrock, Rochester, NY) before detection with an Andor iXon 897 EMCCD (Oxford Instruments, Abingdon, UK).

Samples were placed in a sealed N₂-purged temperature stage (THMS600, Linkam, Tadworth, Surrey, UK) attached to the microscope stage. The imaging beam was sent through the bottom observation window of the temperature stage, while the bleaching beam was sent through the top observation window. The temperature was varied from 20 to 100 °C, with measurements conducted every 20 °C. The temperature stage has an accuracy of 0.01 °C. FRAP movies were collected after a 30-minute temperature equilibration time. FRAP bleaching parameters, acquisition time, and total frames collected were adjusted to fit each temperature step (Table S1).

FRAP Data Analysis. FRAP images were analyzed using ImageJ 2.1.0. Fluorescence intensities were measured for every image in the movie in two regions, the photobleached region and a control region outside the illumination of the laser used for photobleaching. The intensities in the photobleached region were normalized to the intensities in the control region. The normalized data were loaded into MATLAB 2021A for curve fitting using the custom fit tool with the anomalous diffusion model (eq 1).^{75,76}

$$F(t) = \frac{F_0 + F_{\text{in}} \left(\frac{t}{t_{1/2}} \right)^{\alpha}}{1 + \left(\frac{t}{t_{1/2}} \right)^{\alpha}} \quad (1)$$

The parameter F_0 is the initial fluorescence intensity after photobleaching; F_{in} is the fluorescence intensity at an infinite recovery time; $t_{1/2}$ is the time for 50% of the fluorescence to recover; and α is the time exponent providing a measure of Brownian diffusion ($\alpha = 1$). The diffusion coefficient (D) is calculated with

$$D = \frac{r^2}{4t_{1/2}} \quad (2)$$

where r is the radius of the bleached area measured at the $1/e^2$ height of the laser profile.

Raman Spectroscopy. Temperature-dependent Raman spectra were collected using an XploRA Plus Raman confocal upright microscope (HORIBA Scientific, Edison, NJ) equipped with a 785-nm solid-state diode laser operating at 47.81 mW. Raman data were collected in the epi-direction with an Olympus objective (20× magnification, 0.4 numerical aperture). A 100 μm confocal pinhole, 300 μm monochromator slit, and 1200 gr/mm grating was utilized supplying a spectral resolution of approximately 1 cm^{-1} /pixel. The acquisition time was 30 s with 3 accumulations averaged for each measurement, and a total of three measurements were collected at each temperature step. The temperature of the samples was controlled using the same temperature stage used for FRAP measurements. In addition to the three aforementioned sample geometries, Raman spectra were measured in the bulk solvent using the sample preparation previously described.⁷⁷ Igor Pro ver. 6.37 (WaveMetrics, Inc., Lake Oswego, OR) scientific analysis and graphing software was used to analyze the Raman spectra. Spectral peaks were fit with a Gaussian function after baseline correction to measure peak location and full-width-at-half-maximum.

RESULTS AND DISCUSSION

The temperature-dependent viscosity was collected for bulk (i.e., free from the influence of substrates) glycerol, [C₆mim][Cl], and [C₈mim][Cl] since the solvent viscosity affects diffusion properties. Glycerol was included in our studies, as it is a Newtonian fluid with similar viscosities to the ILs, and it does not exhibit nanoscale structural heterogeneities. As with ILs, glycerol exhibits temperature-dependent hydrogen bonding; with increasing temperature, there is a decrease in the overall strength of the hydrogen bonding network.^{78,79} Consistent with other reports, there is a non-linear relationship between the inverse of viscosity and the temperature of these solvents (Figure 3). Also consistent with other reports, an

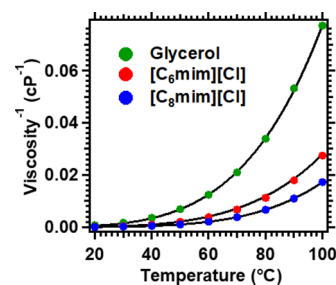


Figure 3. Viscosities of glycerol, [C₆mim][Cl], and [C₈mim][Cl] measured from 20–100 °C measured at increments of 10 °C. All samples were dried under vacuum at room temperature until the water content was below 100 ppm. Solid black lines represent the fit to eq 3. Error bars are not visible on this scale ($n = 3$).

increase in alkyl chain length from [C₆mim][Cl] to [C₈mim][Cl] is associated with an increase in viscosity at all temperatures.⁸⁰ Differences in measured values from this work and previously cited work may come from many sources including: impurities, instrumentation, temperature uncertainties, unreported experimental conditions, and others.^{81–84} In our study, all solvents were dried to obtain a water content of less than 100 ppm. Comparisons between the viscosities reported in this work and previous reports are compiled in Table S3. The inverse of viscosity as a function of temperature was fit to eq 3, derived by Gahtee et al.^{85–87}

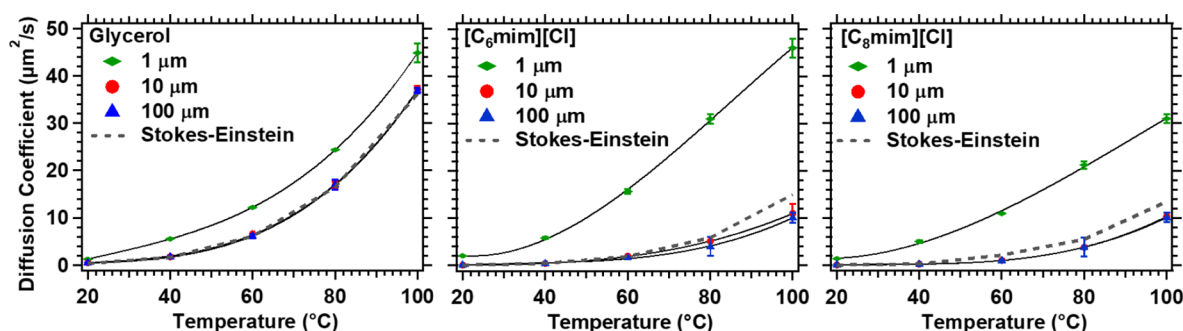
$$\frac{1}{\eta} = (a + bT)^{1/\phi} \quad (3)$$

where η represents viscosity, and a , b , and ϕ are material-dependent constants. The equation applies the power law for viscosity of ILs as a function of temperature with mode coupling theory.^{86,87} Values of a , b , and ϕ from the fits to data in Figure 3 are listed in Table S2.

Using FRAP, the effects of temperature and film thickness on the diffusion properties of ATTO 647 were monitored in glycerol, [C₆mim][Cl], and [C₈mim][Cl]. As a Newtonian fluid, diffusion properties within glycerol should follow the Stokes–Einstein relationship with an $\alpha \approx 1$ for the bulk solvent; therefore, glycerol acts as our control measurement. A representative FRAP curve for each solvent, temperature, and film thickness is found in Figure S3. FRAP data collected as an additional control in the absence of fluorophore are plotted in Figure S4; these curves indicate that in the absence of fluorophore, no bleaching event is detected in the pure solvent. As expected, the diffusion coefficient increases (i.e., there is a decrease in $t_{1/2}$) with increasing temperature and decreasing viscosity.

Table 2. Temperature-Dependent Diffusion Properties for Glycerol, [C₆mim][Cl], and [C₈mim][Cl] Films with 1, 10, and 100 μm Thickness

	temp. (°C)	1 μm		10 μm		100 μm	
		diffusion coefficient ($\mu\text{m}^2/\text{s}$)	α	diffusion coefficient ($\mu\text{m}^2/\text{s}$)	α	diffusion coefficient ($\mu\text{m}^2/\text{s}$)	α
glycerol	20	1.23 $\pm$ 0.04	0.91 $\pm$ 0.03	0.49 $\pm$ 0.01	0.91 $\pm$ 0.03	0.43 $\pm$ 0.02	0.89 $\pm$ 0.01
	40	5.396 $\pm$ 0.008	0.929 $\pm$ 0.008	1.73 $\pm$ 0.03	0.912 $\pm$ 0.006	1.73 $\pm$ 0.05	0.915 $\pm$ 0.005
	60	12.2 $\pm$ 0.05	1.005 $\pm$ 0.003	6.6 $\pm$ 0.02	0.92 $\pm$ 0.02	6.1 $\pm$ 0.4	0.93 $\pm$ 0.02
	80	24.38 $\pm$ 0.05	1.008 $\pm$ 0.006	16.9 $\pm$ 0.7	0.97 $\pm$ 0.01	17 $\pm$ 1	0.955 $\pm$ 0.006
	100	45 $\pm$ 2	1.01 $\pm$ 0.01	37.2 $\pm$ 0.7	1.00 $\pm$ 0.02	36.9 $\pm$ 0.6	0.98 $\pm$ 0.01
[C ₆ mim] [Cl]	20	2.02 $\pm$ 0.02	0.883 $\pm$ 0.009	0.090 $\pm$ 0.009	0.88 $\pm$ 0.01	0.09 $\pm$ 0.04	0.88 $\pm$ 0.02
	40	5.8 $\pm$ 0.2	0.96 $\pm$ 0.04	0.43 $\pm$ 0.04	0.88 $\pm$ 0.01	0.44 $\pm$ 0.02	0.886 $\pm$ 0.005
	60	15.6 $\pm$ 0.5	1.006 $\pm$ 0.003	1.97 $\pm$ 0.04	0.927 $\pm$ 0.005	1.63 $\pm$ 0.02	0.92 $\pm$ 0.02
	80	31 $\pm$ 1	1.008 $\pm$ 0.006	5.1 $\pm$ 0.02	0.93 $\pm$ 0.09	4 $\pm$ 2	0.93 $\pm$ 0.02
	100	46 $\pm$ 2	1.008 $\pm$ 0.006	11 $\pm$ 2	0.99 $\pm$ 0.01	10 $\pm$ 1	0.99 $\pm$ 0.01
[C ₈ mim] [Cl]	20	1.35 $\pm$ 0.04	0.84 $\pm$ 0.02	0.023 $\pm$ 0.004	0.81 $\pm$ 0.01	0.021 $\pm$ 0.002	0.82 $\pm$ 0.01
	40	5.0 $\pm$ 0.3	0.90 $\pm$ 0.01	0.19 $\pm$ 0.02	0.845 $\pm$ 0.005	0.19 $\pm$ 0.02	0.84 $\pm$ 0.01
	60	10.9 $\pm$ 0.1	1.005 $\pm$ 0.004	1.02 $\pm$ 0.02	0.862 $\pm$ 0.009	0.97 $\pm$ 0.04	0.85 $\pm$ 0.02
	80	21.2 $\pm$ 0.8	1.005 $\pm$ 0.009	3.77 $\pm$ 0.05	0.93 $\pm$ 0.03	3.8 $\pm$ 0.2	0.92 $\pm$ 0.02
	100	31.0 $\pm$ 0.9	1.009 $\pm$ 0.001	10.3 $\pm$ 0.1	0.990 $\pm$ 0.007	10.1 $\pm$ 0.2	0.98 $\pm$ 0.05

**Figure 4.** Diffusion coefficient as a function of temperature for the three solvents and thicknesses. Solid black lines represent fit to a polynomial equation for visual guidance. The dotted line is a calculation using the Stokes–Einstein equation calculated with the measured viscosity for each solvent at the indicated temperature as well as an approximate fluorophore hydrodynamic radius of 5.0 Å. Average and standard deviation of the three measurements.

From the fits using eqs 1 and 2 of the FRAP curves, the diffusion coefficient and α values were extracted for each solvent and temperature (Table 2). Across all solvents, there is a non-linear increase in the diffusion coefficient with increasing temperature (Figure 4), which is expected with a non-linear increase in inverse of viscosity with increasing temperature (Figure 3).

For glycerol, the 10 and 100 μm films exhibit a temperature-dependent diffusion coefficient that follows the Stokes–Einstein relationship (Figure 4). For the same thicknesses, the [C₆mim][Cl] and [C₈mim][Cl] films exhibit a negative deviation from the Stokes–Einstein relationship. Uncertainties in the temperature and viscosity measurements, as well as reasonable changes in the size of the fluorophore, may account for the deviations when comparing our experimental data to values calculated using the Stokes–Einstein equation. A value of 5.0 Å was chosen as the hydrodynamic radius of ATTO 647 to calculate the diffusion coefficient using the Stokes–Einstein equation. This value is an approximate value of many similar fluorophores,^{88–90} particularly those measured in ILs.⁹¹ It is also possible that solute diffusion–viscosity decoupling contributes to the negative deviation.⁹²

For all solvents, the 1 μm films reveal a positive deviation from the Stokes–Einstein relationship; the magnitude of this deviation is larger for [C₆mim][Cl] and [C₈mim][Cl] ILs than

for glycerol, particularly at higher temperatures. The axial resolution of the FRAP instrument, as defined by Abbe,⁹³ is approximately 14 μm . Thus, the entirety of the 1 and 10 μm samples are within the signal collection volume, while approximately 14% of the 100 μm sample is within the collection volume. Differences in the data for the 1 and 10 μm films are not likely due to the differences in how the films were sampled. It is likely that a small fraction of the fluorophores interact with the glass substrate, which will affect their diffusion properties. Molecules influenced by the glass surface are expected to diffuse slower than those in the bulk solution.⁹⁴ Therefore, if surface-adsorbed molecules were contributing significantly to the FRAP signal, we would expect the diffusion coefficient to be the slowest in the 1 μm film. This is opposite to what we measured; thus, we conclude that the surface-adsorbed molecules negligibly influence the reported diffusion coefficients. Since the samples were prepared and sealed in a dry N₂-filled glove box, it is unlikely that the cause of this positive deviation is from water sorption. (It is possible the sample thickness decreases with temperature; this does not alter our conclusions). As previously mentioned, Filippov et al. noted increased self-diffusion coefficients in micro-confined (~ 3 to 4 μm) ILs compared with that of the bulk IL.^{51,52} They also noted an increase in the diffusion coefficient of lithium ions in micro-confined ethylammonium nitrate compared with

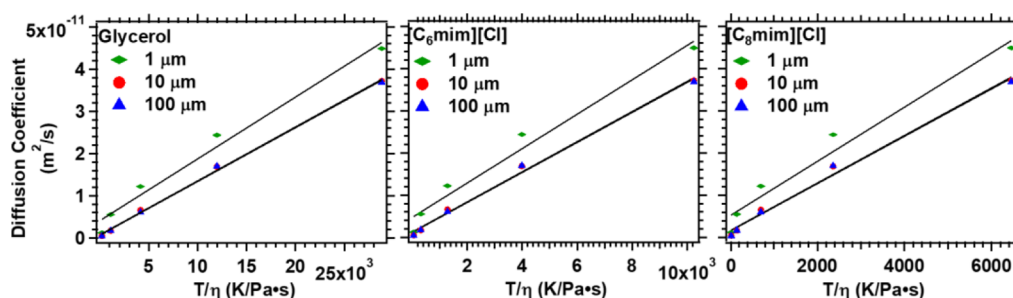


Figure 5. Diffusion coefficient as a function of T/η , where T represents temperature and η represents the viscosity of the bulk solution for the three solvents and sample thicknesses. Solid black lines represent a linear fit for visual guidance. Average and standard deviation of three measurements.

that of the bulk IL.⁹⁵ Consistent with other reports, a possible cause of the positive deviation in the diffusion coefficient for the 1 μm IL films is a decrease in solvent viscosity near the substrate due to alterations in the intermolecular interactions.^{48,51–53,95–98} Figure 5 shows the diffusion coefficient of ATTO 647 plotted with the ratio of temperature to the viscosity of the bulk solvent. The 1 μm films show the most non-linear dependence of the diffusion coefficient on the ratio of temperature to viscosity; the bulk solvent viscosity may not be an acceptable estimate for the 1 μm films.

Changes in intermolecular interactions within glycerol, $[\text{C}_6\text{mim}][\text{Cl}]$, and $[\text{C}_8\text{mim}][\text{Cl}]$ could be influential in altering the temperature-dependent diffusion of ATTO 647 when the solvent is in the proximity of solid substrates. To investigate the possible altered intermolecular interactions, temperature-dependent Raman spectra were collected for 10 and 100 μm films as well as the bulk solvent (Figure S5). There are shifts in peaks assigned to functional groups that may participate in hydrogen bonding when comparing Raman spectra collected at 20 and 150 $^{\circ}\text{C}$ (Figure S5 and Table S3), showing an expected trend of decreasing hydrogen bonding strength with increasing temperatures for these solvents.

For all solvents and all film thicknesses, there is an increase in the α parameter, or an increase in Brownian diffusion, with increasing temperature (Table 2). With decreasing temperature, $\alpha < 1$, indicating constrained diffusion. Constrained diffusion in ILs is reported to be caused by nanoscale structural heterogeneity as predicted by computational studies and previous experimental measurements.^{27,83} These nanoscale structural heterogeneities are caused by the association of polar and non-polar moieties that exist over longer lifetimes than observed in molecular solvents.^{56,99,100} The nanoscale structural heterogeneities of ILs become less pronounced, as segregation of non-polar domains decreases with increasing temperature.^{101,102} It is known that the nanoscale structural heterogeneities are larger with alkylimidazolium halide ILs with longer alkyl chains.¹⁰³ The differences in the nanoscale structural heterogeneities are relatively small when comparing $[\text{C}_6\text{mim}][\text{Cl}]$ and $[\text{C}_8\text{mim}][\text{Cl}]$; an increase of two $-\text{CH}_2-$ units leads to a 0.4 nm increase in the size of the non-polar domain.¹⁶ The lower α values (Table 2) at temperatures below 60 $^{\circ}\text{C}$ are consistent with more pronounced nanoscale heterogeneity in $[\text{C}_8\text{mim}][\text{Cl}]$ compared with $[\text{C}_6\text{mim}][\text{Cl}]$.^{16,104}

For all three solvents, Brownian diffusion ($\alpha = 1$) is measured at temperatures above 60 $^{\circ}\text{C}$ for the 1 μm films and above 100 $^{\circ}\text{C}$ for the 10 and 100 μm films. The nanoscale heterogeneity, as measured by FRAP, is disrupted at a similar temperature for both $[\text{C}_8\text{mim}][\text{Cl}]$ and $[\text{C}_6\text{mim}][\text{Cl}]$ in 1 μm

films. Also, a higher temperature is required to disrupt this nanoscale ordering for both ILs in 10 and 100 μm films. The α values were expected to be 1 for glycerol, which would be consistent with our previous measurements as well as a lack of expected nanoscale organization.^{105,106} In the glycerol films, deviations from the expected $\alpha = 1$ could be due to the strong three-dimensional hydrogen bonding network that induces aggregate structures.¹⁰⁶ With increasing temperature, the strength of this hydrogen bonding network decreases, thereby allowing for Brownian motion of ATTO 647.

CONCLUSIONS

FRAP was utilized to determine the temperature-dependent diffusion coefficients and constrained diffusion behavior of ATTO 647 in glycerol, $[\text{C}_6\text{mim}][\text{Cl}]$, and $[\text{C}_8\text{mim}][\text{Cl}]$. The observed temperature-dependent diffusion coefficients and constrained diffusion behavior can be attributed in part to the altered viscosities and nanoscale ordering compared to the bulk solvent (i.e., located far enough away from a surface so as not to be influenced by it). Comparing the values of the diffusion coefficient with temperature, $[\text{C}_6\text{mim}][\text{Cl}]$, and $[\text{C}_8\text{mim}][\text{Cl}]$ have a similar behavior, whereas when comparing the α values, $[\text{C}_6\text{mim}][\text{Cl}]$ is more similar to glycerol, a solvent that lacks the nanoscale structure when it is not in the proximity of a substrate. For all solvents, the diffusion properties within the 1 μm films are different compared to the 10 and 100 μm films. These results provide a new experimental understanding of the physiochemical complexity of alkylimidazolium-based ILs as a function of temperature and micron-scale confinement and will be a useful foundation for understanding how these properties may affect a variety of separation systems using these solvents.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.2c01588>.

Sample preparation, absorption spectra of solutions and fluorophore, FRAP collection parameters, viscosity fit values, representative FRAP curves, and Raman spectral assignments (PDF)

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Notes

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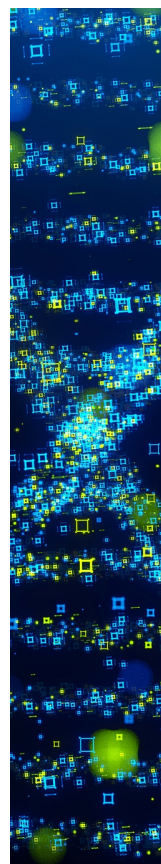
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